MIL-I-3606E
30 October 1980
SUPERSEDING
MIL-I-3606D
7 July 1975

MILITARY SPECIFICATION

INK, PRINTING: LITHOGRAPHIC OFFSET, WITH DRIER: GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lithographic offset printing ink compounded with separate drier, to be mixed together at the time of use. This is to accelerate the drying time and prolong storage life. These inks are designed for overprinting; and, except for black, are transparent type.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 7510

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

UU-P-465	- Paper, Book.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-96	- Cans, Metal, 28 Gage and Lighter.
PPP-T-66	- Tape, Packaging, Vinyl Plastic Film.

MILITARY

(See Supplement 1 for list of associated specification sheets.)

STANDARDS

FEDERAL

FED-STD-313	- Preparation and Submission of Material Safety Data Sheets.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables For Inspection by Attributes.
MIL-STD-129	- Marking For Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

GRAPHIC ARTS TECHNICAL FOUNDATION (GATF)

No. 502/4 - Offset Lithographic Platemaking.

(Application for copies should be addressed to the Graphic Arts Technical Foundation, 4615 Forbes Avenue, Pittsburgh, PA 15213.)

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NATIONAL PRINTING INK RESEARCH INSTITUTE (NPIRI)

Standard Test Methods.

(Application for copies should be addressed to the National Printing Ink Research Institute, Francis MacDonald Sinclair Memorial Laboratory, Lehigh University, Building 7, Bethlehem, PA 18015.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1729 - Visual Evaluation of Color Differences of Opaque Materials.
D 2244 - Instrumental Evaluation of Color Differences of Opaque Materials.
G 27 - Operating Xenon-Arc Type Apparatus For Light Exposure of Nonmetallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The ink shall be compounded for lithographic offset printing on sheet-fed presses. The drier shall be separate. The ink shall be formulated to permit its use directly from the container with only the drier being added. The ink shall be compatible with similar inks when overprinted thereon, and shall be suitable for both line and halftone reproduction.

3.1.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Standard sample. A standard wet sample, press specimen, or set of drawdowns, at the option of the Government (see 6.2), will be furnished by the contracting officer (see 6.3). When a wet sample is

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furnished, the contractor shall prepare standard drawdowns on the same paper and with the same film applicator as that used to prepare test drawdowns (see 4.4.2.1).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free from chemicals or compounds that will react injuriously with other materials contacted in the lithographic process and shall be free from other impurities that may affect the serviceability of the product, or that would be injurious to operating personnel.

3.4 Storage temperature. The packaged ink and supplementary drier shall show no signs of gelling, livering, change in viscosity, or any deterioration by storage under the following conditions:

Air temperature:
High - Plus 160°F (plus 71° C) for 4 hours.
Low - Minus 30°F (minus 34° C) for 12 hours. No solar radiation.

3.5 Composition.

3.5.1 Pigments. The pigments used in the ink shall be insoluble in water and dilute acids.

3.5.2 Vehicles. The vehicles used in the ink shall have wetting properties to protect the pigments against chemical action and bleeding. Wetting agents shall not be used that will detrimentally affect the working properties of the ink by introducing emulsification on the press.

3.5.3 Driers. Black ink shall incorporate a minimal amount of cobalt linoleate, factory-mixed into the ink. Transparent ink shall contain no cobalt drier.

3.6 Quantitative.

3.6.1 Specific gravity. The specific gravity of the ink shall be between 1.10 and 1.26 at a temperature of plus 73° F plus or minus 3.5° F (plus 23° C plus or minus 2° C).

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3.6.2 Drying. Unless otherwise specified (see 6.2), the ink without additional drier shall dry (see 6.4) in 24 hours plus or minus 4 hours at a temperature of plus 73° F plus or minus 3.5° F (plus 23° C plus or minus 2° C), and a relative humidity of 50 percent plus or minus 2 percent. The ink shall dry in not more than 8 hours at a temperature of plus 73° F plus or minus 3.5° F (plus 23° C plus or minus 2° C), and a relative humidity of 50 percent plus or minus 2 percent when the supplementary drier in the amount recommended by the manufacturer is added to the ink.

3.7 Qualitative.

3.7.1 Tinctorial strength. The tinctorial strength of the ink shall match the tinctorial strength of the sample drawdown, plus or minus 10 percent of bleaching white.

3.7.2 Light fastness. Unless otherwise specified (see 6.2), the light fastness of the ink shall be not less than the light fastness of the sample drawdown.

3.8 Color. The color of the ink shall be as specified (see 6.2), and shall match the color of the applicable specification sheet. When a special color is required, the color shall be as specified (see 6.2) and shall match that of the wet sample, specimen, or drawdowns furnished by the contracting officer.

3.9 Printing. The ink shall produce a fine 150- and 300-line highlight dot, and a clean, open 150- and 300-line shadow dot line copy and a solid bar on paper conforming to UU-P-465, Type II.

3.10 Material Safety Data Sheet. A material safety data sheet shall be prepared in accordance with FED-STD-313 for all chemicals used in offset printing process. Special emphasis should be given to proper disposal of chemical wastes.

3.11 Supplementary drier. Unless otherwise specified (see 6.2), the requisite drier shall be furnished separate from the ink, and shall consist of a nonleaded paste. Drier shall be furnished in sufficient quantity for each can of ink to exceed by 50 percent the amount required for 8-hour drying. The drier, when added to the ink in the required proportions recommended by the manufacturer, shall produce an ink compound satisfactory for use on medium and large sheet-fed offset presses.

3.12 Identification marking. Each can of ink shall be clearly and legibly lithographed or marked with water-resistant ink, or furnished with a label printed with water-resistant ink on a good grade of printing paper and firmly affixed to the can with a water-resistant adhesive. The markings shall clearly identify the product and its manufacturer (see 5.3.3).

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3.13 Workmanship. The ink shall be smooth, homogeneous, free from lumps, strings, and objectionable odor. Labels shall be clearly printed and neatly and securely affixed to cans. Ink containers shall show no evidence of spillage or leakage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Unit of product. For the purpose of inspection, a pound of ink and the quantity of drier recommended by the manufacturer for the conditions under which the tests are to be run shall be considered a unit of product.

4.3.2 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Inspection Level S-2.

4.3.3 Examination. Sample units selected in accordance with 4.3.2 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective.

4.3.4 Tests. Sample units selected in accordance with 4.3.2 shall be tested as specified in 4.4.2. AQL shall be 4.0 percent defective.

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4.4 Inspection procedure.

4.4.1 Examination. The ink and the drier shall be examined as specified herein for the following defects:

101. Material not as specified.
102. Material Safety Data Sheet missing.
103. Supplementary drier not furnished with the ink.
104. Identification marking missing, incomplete, or illegible.
105. Workmanship not as specified.

4.4.2 Tests. All tests shall be made on the same run or batch of paper.

4.4.2.1 Drawdowns. Test drawdowns for the preparation of color masstone, tinctorial strength, and drying time test shall be made using a 0.0005 Bird film applicator. Test drawdowns for the preparation of color undertone shall be made using a flat, straight-edged instrument not less than 2-inches in width. Drawdowns shall be on paper conforming to UU-P-465, Type II.

4.4.2.2 Specific gravity. Determine the specific gravity of ink at a temperature of plus 73° F plus or minus 3.5° F (plus 23° C plus or minus 2° C) by NPIRI Standard Test Method D-2 or its equivalent. A specific gravity of less than 1.10 or more than 1.26 shall constitute failure of this test.

4.4.2.3 Drying time. Maintain an ambient temperature of 73° F plus or minus 3.5° F (plus 23° C plus or minus 2° C), and a relative humidity of 50 percent plus or minus 2 percent during this test. Determine the drying time of the ink by making a drawdown without adding drier, and another drawdown with the recommended amount of drier. A drying time other than 24 hours, plus or minus 4 hours without drier, or more than 8 hours with drier, shall constitute failure of this test (see 6.4).

4.4.2.4 Tinctorial strength. Bleaching white shall be formulated as specified in the following formula:

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Composition of bleaching white

Ingredient	Amount in grams
Zinc oxide (white seal)	79.03 ±0.05
No. 1 transparent (clear) varnish	19.74 ±0.02
White petroleum	1.23 ±0.01

Determine the tinctorial strength of the ink as follows:

(a) Colors. Mix 5 grams of bleaching white with 100 milligrams (mg) of ink. Prepare a drawdown and immediately compare this drawdown with the standard ink-plus-bleaching white drawdown. Drawdowns having a lesser strength than the standard ink-plus-bleaching white drawdown shall constitute failure of this test. Drawdowns of greater strength shall be retreated as follows. Reduce the specimen by adding 50 mg of bleaching white to 1 gram of the specimen ink-plus-bleaching white mixture. Prepare a new drawdown and immediately compare the new sample drawdown to the standard ink-plus-bleaching white drawdown. Drawdowns having a greater strength than the standard drawdown shall constitute failure of this test.

(b) Blacks. Mix 5 grams of bleaching white with 100 mg of ink. Prepare a drawdown and immediately compare this drawdown with a standard ink-plus-bleaching white drawdown. Drawdowns having a lesser strength than the standard drawdown shall constitute failure of this test.

4.4.2.5 Light fastness. Determine the light fastness of the ink by placing a sample ink drawdown and a standard drawdown in a fadeometer, ASTM G 27, 2500 watt, Method A for 70 hours. A light fastness of the undertones of the sample drawdown less than the light fastness of the undertones of the standard drawdown shall constitute failure of this test.

4.4.2.6 Storage temperature. The packaged can of ink, with its supplementary drier, shall be subjected to the following temperature tests. Any evidence of gelling, livering, a change in viscosity, or any other deterioration shall constitute failure of this test.

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4.4.2.6.1 High temperature. The ink and drier shall be placed in an ambient temperature of plus 160° F (plus 71° C) for not less than four hours, after which they shall be removed, allowed to cool to plus 68° F plus or minus 5° F (20° C plus or minus 3° C), opened and examined.

4.4.2.6.2 Low temperature. The ink and drier shall be placed in an ambient temperature of minus 30° F (minus 34° C) for not less than twelve hours, after which they shall be removed, allowed to warm to plus 68° F plus or minus 5° F (20° C plus or minus 3° C), opened, and examined.

4.4.2.7 Color. Prepare drawdowns in accordance with 4.4.2.1 and allow to dry. Drawdowns and press specimens after the printing test specified in 4.4.2.8 shall be compared in accordance with ASTM D 1729 and D 2244. Comparisons shall be made by persons who have passed a Hardy-Rank Rittler pseudo-isochromatic plate test or equivalent color perception test, or who have been certified by a licensed opthalmologist as having full-range trichromatic vision. Failure of the drawdown or specimen of the ink to match that made from the Government's standard ink shall constitute failure of this test.

4.4.2.8 Printing. Run 5,000 impressions using line copy and 150- to 300-line halftone copy and solid bar on paper conforming to UU-P-465, Type II. The run shall be made on a standard sheet-fed offset press, 17 by 22 inches or larger, operating at a speed of from 4,000 to 6,000 impressions per hour. The plates used shall be aluminum and shall be presensitized either commercially or in accordance with GATF No. 502/4. Examine not less than one of every 100 sheets with a 9-power magnifying glass. Any changes in the size of the dot structure shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

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4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106.	Materials and containers not as specified for Level A or B.	Each incorrect material or container shall be considered one defect.
107.	Ink contained in quantities other than specified for Level A or Commercial.	
108.	Waxed paper not placed on top of ink as specified for Level A.	
109.	Cans not sealed as specified for Level A.	
110.	Drier not contained in quantities required for each can of ink as specified for Level A or Commercial.	
111.	Consolidation not as specified for Level A.	
112.	Fiberboard boxes not closed by Method V as specified for Level A.	
113.	Marking missing, illegible, incorrect, or incomplete for Level A, Level B, or Commercial.	

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Ink. The ink shall be contained in 1- or 5-pound metal cans, as specified (see 6.2), conforming to PPP-C-96, Type VI. Exterior coating shall be Plan B. Before placing the cover on the can, a sheet of waxed paper of the same dimension as the interior circumference of the can shall be placed flat on top of the ink. The covers shall be installed and the can sealed by the application of 1-inch wide tape conforming to PPP-T-66, Type I, class optional, applied around the can and over the lip of the cover.

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5.1.1.2 Supplementary drier. The supplementary drier shall be contained in the quantity required for each can of ink in a commercial-type, collapsible metal or plastic tube with screwcap, or in a metal can conforming to PPP-C-96, Type V, Class 1 or 2, or Type VI. Type VI cans shall be sealed as specified in 5.1.1.1, when applicable.

5.1.1.3 Consolidation.

5.1.1.3.1 One pound cans of ink. Ink contained in 1-pound cans and the required tubes or cans of drier shall be consolidated together in quantities of 48 each, in a close-fitting box conforming to PPP-B-636, W5c, style optional. The box shall be provided with fiberboard separators and pads. The pads shall be placed between layers and on the top and bottom. Separators and pads shall be of the same material as the box. The box shall be closed by Method V of appendix to the box specification.

5.1.1.3.2 Five-pound cans of ink. Ink contained in 5-pound cans and the required tubes or cans of drier shall be consolidated together as specified in 5.1.1.3.1 except the quantities shall be six each.

5.1.2 Commercial. The ink and drier in the quantities specified in 5.1.1.1 and 5.1.1.2 shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. Ink with the required drier, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional or PPP-B-621, Class 2, style optional. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Ink with the required drier preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, Domestic Type; PPP-B-621, Class 1; PPP-B-640, Class 2, style optional; or PPP-B-636, V3c. The gross weight of each box shall not exceed the weight or size limitations of the applicable box specification.

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5.2.3 Commercial. Ink with the required drier, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (Level A and Level B) shall be in accordance with MIL-STD-129 and the additional marking specified in 5.3.3.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188 and the additional marking specified in 5.3.3.

5.3.3 Additional marking. In addition to the marking specified in 5.3.1 or 5.3.2, the date of manufacture and the recommended amount of drier to be added under normal conditions for not more than 8 hours drying time shall be marked on the cans. (Normal conditions are 73° plus or minus 3.5° F (23° plus or minus 2° C) at 50 percent relative humidity.)

6. NOTES

6.1 Intended use. The ink with drier is intended for use on lithographic printing presses operated in field units of the Armed Forces under various climatic conditions. This ink is intended primarily for printing topographic maps.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of applicable specification sheet (see 3.1.1).
(c) Type of standard sample to be supplied by the government (see 3.2).
(d) Drying characteristics of ink if other than as specified (see 3.6.2).
(e) Light fastness required if other than as specified (see 3.7.2).
(f) Color required (see 3.8).
(g) Color, if a special color is required (see 3.8).
(h) When supplementary drier is not required (see 3.11).
(i) When certification may be accepted in lieu of the test (see 4.4.2.6.3).
(j) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(k) Size of cans required (see 5.1.1.1).

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6.3 Samples. The contracting officer should procure the necessary quantity of specimens to accompany the IFB or RFP. A set of drawdowns consists of masstone, a tinctorial strength, and an undertone or a press specimen. The contracting officer should procure wet samples for the successful bidders. Specimens and samples of the colors issued for Army troop use may be obtained only by contracting officers from the Defense Mapping Agency Topographic Center, Department of Graphic Arts, Washington, DC 20315.

6.4 Drying time. Drying time shall be defined as the time lapse between the preparation of the drawdown and when a 3 to 4 pound finger pressure on the surface of the drawdown no longer leaves an impression.

Custodian:
Army - ME

Review activity:
Army - GL

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 7510-0313

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